Arunima Anand

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EDUCATION

M.S. in Data Science | University of California, San Diego — Sep 2025 – Dec 2026 | San Diego, CA
4.00 GPA
Relevant Coursework: Machine Learning, Recommender Systems, ML Systems, Data-Driven Text Mining, Scalable Data Systems

Bachelor's in Economics (Minor in Mathematics) | University of Delhi — Jul 2019 – Jul 2022 | Delhi, India
3.88 GPA (First Division with Honors)

PROFESSIONAL EXPERIENCE

Applied Science Intern (Measurement & Identity) | The Trade Desk — Jun 2026 – Sep 2026 | New York, NY

- Developed a prototype **probabilistic audience-reach estimator** over **16B+ ad events** using censored frequency modeling, identity-graph signals, HyperLogLog sketches, and vectorized feature embeddings for cohorts with limited identity coverage.
- Built **PySpark pipelines** across **multiple aggregation levels** (ad group, campaign, and advertiser) and validated reach estimates against **two independent signals** using temporal and entity holdouts, sensitivity tests, and A/B tests.
- Applied **knowledge distillation** to transfer signal from richer representations into LightGBM models using production-efficient features, enabling **sub-50 ms inference at scale**.
- Productionized estimator outputs through **Databricks APIs integrated** with the **trading platform**, powering a user-facing dashboard with reach analysis, cohort diagnostics, and continuous model monitoring.

Data Scientist II (AI/ML Engineering) | Deloitte — Oct 2022 – Aug 2025 | Delhi, India

- Developed and benchmarked an **automated demand forecasting platform across 7,000+ part-level time series**, engineering lag and rolling-window features, evaluating **Prophet, XGBoost, LSTM, and ARIMA**, and selecting the best-fit model per series**,** achieving **93.63% forecast accuracy and $5M+ in savings**.
- Productionized **training and batch inference workflows** on AWS SageMaker, S3, and Step Functions with automated feature generation, retraining, validation, and drift monitoring, reducing **end-to-end runtime** by **91%**.
- Designed and deployed an **explainable, configurable ML framework** for **product scoring and classification** using feature-weighted aggregation, **distribution-aware binning, Random Forest,** and **LightGBM**, achieving **95% holdout accuracy** and maintaining production stability through drift-aware recalibration and model versioning.
- Architected an **NLP pipeline** to **standardize** inconsistent **manual defect descriptions** and surface **recurring defect patterns** using **BERT embeddings** and **HDBSCAN clustering**, mapping **31%** of records to canonical labels and improving detection yield by **1.6×**.

PROJECTS & RESEARCH

Personalized Ad Ranking for a Housing Marketplace — Oct 2025 – Dec 2025

- Built a **personalized ad** and **listing ranking system** over **500K+ clickstream events** and **100K+ user profiles**, using browsing behavior and user-item affinity to rank relevant property listings and campaign content for each user.
- Generated **user-level targeting scores** to **identify high-affinity campaign audiences**, improving planning signal quality by **32%**.

Evidence-Grounded LLM Evaluation & Retrieval — Jan 2026 – Mar 2026

- Designed a distilled **LLM-as-a-judge** with **source-evidence retrieval**, using **PyTorch** and **LoRA** to adapt **Qwen2.5-3B** for summary evaluation across coherence, consistency, fluency, and relevance.
- Conducted fine-grained meta-evaluation against **human annotations**, achieving **75.5% off-by-1 agreement** and identifying a **ranking-calibration tradeoff** in automated evaluation.

Computer Vision + LLM Food Waste Intelligence Platform — Apr 2026

- Built a **CV** and **LLM platform** that converts restaurant **food-waste images** into **structured waste records**, using S3, Lambda, AWS Rekognition, Grok Vision, and Databricks for ingestion, deduplication, classification, and storage.
- Served **waste trends** and **Prophet-based weekly forecasts** through **FastAPI** and **React** to support restaurant waste monitoring and inventory planning.

SKILLS

- **Programming**: Python, SQL, Java, C++, R
- **Libraries & Frameworks:** PyTorch, TensorFlow, PySpark, scikit-learn, NumPy, Pandas, statsmodels, Matplotlib, NLTK
- **Data Science & Machine Learning:** Recommender systems, ranking, classification, regression, regularization, feature engineering, cross-validation, time-series forecasting, causal inference, A/B testing, clustering, Random Forest, XGBoost, LightGBM
- **Deep Learning, Retrieval & GenAI:** Knowledge distillation, embeddings, LSTMs, transformers, BERT, information retrieval, TF-IDF, semantic search, LLM evaluation, RAG, prompt engineering
- **MLOps, Cloud & AI Engineering:** Databricks, AWS (S3, Lambda, SageMaker), FastAPI, GitHub, CI/CD, real-time model serving, model monitoring and versioning, drift detection, distributed computing, AI-assisted coding and development workflows